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$$w = e^{-x^2-y^2} \text{ met } x = t \text{ en } y = \sqrt{t}$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial w}{\partial x} = -2xe^{-x^2-y^2}$$

$$\frac{\partial w}{\partial y} = -2ye^{-x^2-y^2}$$

$$\frac{dx}{dt} = 1$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}}$$

$$\frac{\partial w}{\partial t} = \left(-2xe^{-x^2-y^2}\right) \cdot 1 + \left(-2ye^{-x^2-y^2}\right) \cdot \frac{1}{2\sqrt{t}}$$

$$= -2xe^{-x^2-y^2} - t^{-\frac{1}{2}}ye^{-x^2-y^2}$$

$$= -2te^{-t^2-t} - t^{-\frac{1}{2}}t^{\frac{1}{2}}e^{-t^2-t}$$

$$= -2te^{-t^2-t} - e^{-t^2-t}$$

$$= (-2t - 1)e^{-t^2-t}$$

References